DR. ANIKA SHARMA

Scientist E — Atmospheric Instrumentation & Applied AI

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PROFESSIONAL SUMMARY

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

AREAS OF EXPERTISE

R&D in avionics systems · 12 years

Twelve years of mission-computer and sensor integration, interface testing (MIL-STD-1553, ARINC 429) and environmental qualification for airborne platforms at Halcyon Avionics Design Bureau.

Edge AI for aerial imagery · 5 years

Research on super-resolution and model compression (pruning, knowledge distillation) for real-time deployment on NVIDIA Jetson devices; publications and a deployed telemetry prototype.

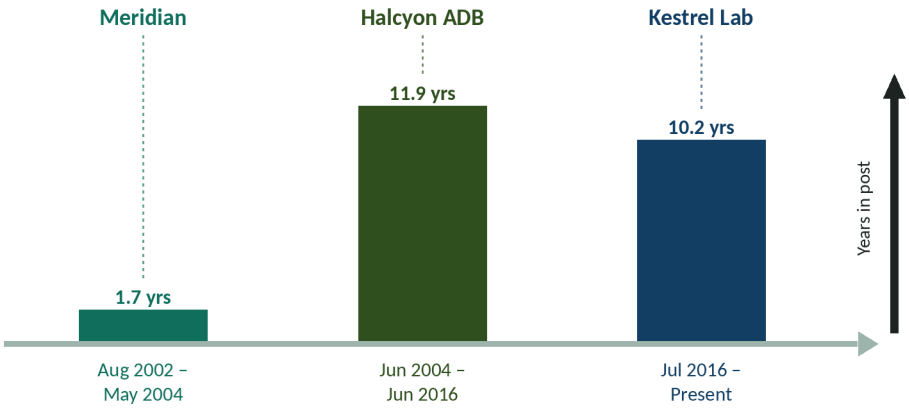
Upper-air instrumentation · 10 years

Planning, testing and maintenance of radiosonde ground systems and upper-air instruments; division workshop management; technical evaluation of procurements.

IoT monitoring systems · 4 years

Design of MQTT, Node-RED, InfluxDB and Grafana pipelines for monitoring ground-station equipment.

CAREER TIMELINE



- Meridian Electronics Limited — Graduate Engineer (Aug 2002 – May 2004): Automated test equipment for avionics units
- Halcyon Avionics Design Bureau — Senior Design Engineer (Jun 2004 – Jun 2016): Mission-computer integration and environmental qualification
- Kestrel Atmospheric Instrumentation Laboratory — Scientist E (Jul 2016 – Present): Upper-air instruments, workshop, IoT and edge-AI programmes

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PROFESSIONAL EXPERIENCE

Scientist E — Jul 2016 – Present

Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi

Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

- ➢ Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.

CERTIFICATIONS & TRAINING

Deep Learning for Edge Devices (40-hour course)
Example Academy of Applied Computing 2022

CORE COMPETENCY

Avionics

- MIL-STD-1553 / ARINC 429
- Avionics integration

Scientific instrumentation

- Environmental qualification testing
- Sensor calibration
- Radiosonde and upper-air systems

Embedded systems

- Embedded C
- ESP32 / STM32 microcontrollers

Computer vision

- Image super-resolution
- Object detection

Bars show my own assessment of proficiency (five steps; five = expert).

LANGUAGES

English
Hindi

Fictional sample profile — not a real person

- ➢ Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- ➢ Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop — Apr 2019 – Present
Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** — Jan 2021 – Present

Senior Design Engineer — Jun 2004 – Jun 2016
Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru
Integration and qualification testing of airborne electronics and UAV payloads.

- ➢ Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.

Graduate Engineer — Aug 2002 – May 2004
Test Systems Group, Meridian Electronics Limited, Bengaluru
Built automated test equipment for avionics line-replaceable units.

- ➢ Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

PROJECTS EXECUTED

Edge super-resolution pipeline for UAV imagery *(Ongoing)* — Sep 2021 – Present
Principal investigator, Kestrel Atmospheric Instrumentation Laboratory
Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

- ➢ Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.

EDUCATION

Ph.D., Electrical and Electronics Engineering — 2021 – ongoing
Northfield Institute of Science and Technology
Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

M.Tech., Avionics — 2010
Coastal Institute of Aeronautical Engineering, Chennai
Grade: CGPA 8.6/10
Thesis: Fault-tolerant data acquisition for airborne test instrumentation

B.E., Electronics and Communication Engineering — 2004
Northfield Institute of Science and Technology, Northfield University, Pune
Grade: First Class with Distinction

PUBLICATIONS

[1] **A. Sharma** and R. Iyer, “Knowledge-distilled lightweight networks for drone image super-resolution,” accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, “Benchmarking embedded GPUs for real-time image enhancement,” Example Symposium on Applied AI, pp. 88-93, Jan 2025.

LEADERSHIP & RESPONSIBILITIES

Supervisor, M.Tech. interns in applied machine learning — 2022 – Present
Kestrel Atmospheric Instrumentation Laboratory
Supervise summer and dissertation interns on edge-AI projects.

Chair, Technical Evaluation Committee for radiosonde procurement — Apr 2023 – Feb 2024
Kestrel Atmospheric Instrumentation Laboratory
Chaired technical evaluation of bids for upper-air sounding systems.

- ➢ Evaluation report accepted by the competent authority.

AWARDS & HONOURS

Best Paper Award — 2023
Example Conference on Instrumentation (ECI 2023)

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